

Course content and style

International Marketing Analytics



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[Office hours: book online \(Google Calendar\)](#)

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The journey begins





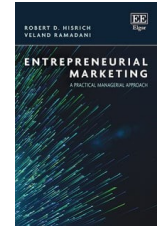
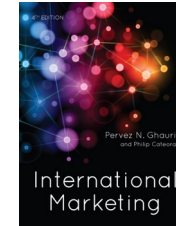
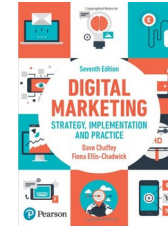
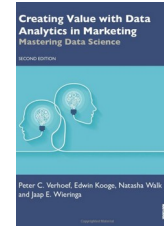
Personal and contact information

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Module Convenor - Dr Arne Floh



- Work Experience
 - International & Industry Background
 - Teaching Experience > 25 Years
- Teaching
 - International Marketing
 - International Business Projects
 - Marketing Research and Marketing Analytics
 - Global Branding
 - Supervision (BSc, MSc, PhD)
- Research
 - Dynamic Capabilities and Firm Performance
 - GenAI and Marketing
 - SMEs and Internationalization
- Private
 - Football
 - Electronic Gadgets
 - Austrian Wine





Marketing analytics is the measurement, management and analysis of marketing performance data to improve marketing effectiveness and return on investment. It turns data into decisions and shows where the next euro should go.



Key questions: how do price, place, product features and promotion drive performance measures such as sales, and how should a fixed budget be split across countries and channels?

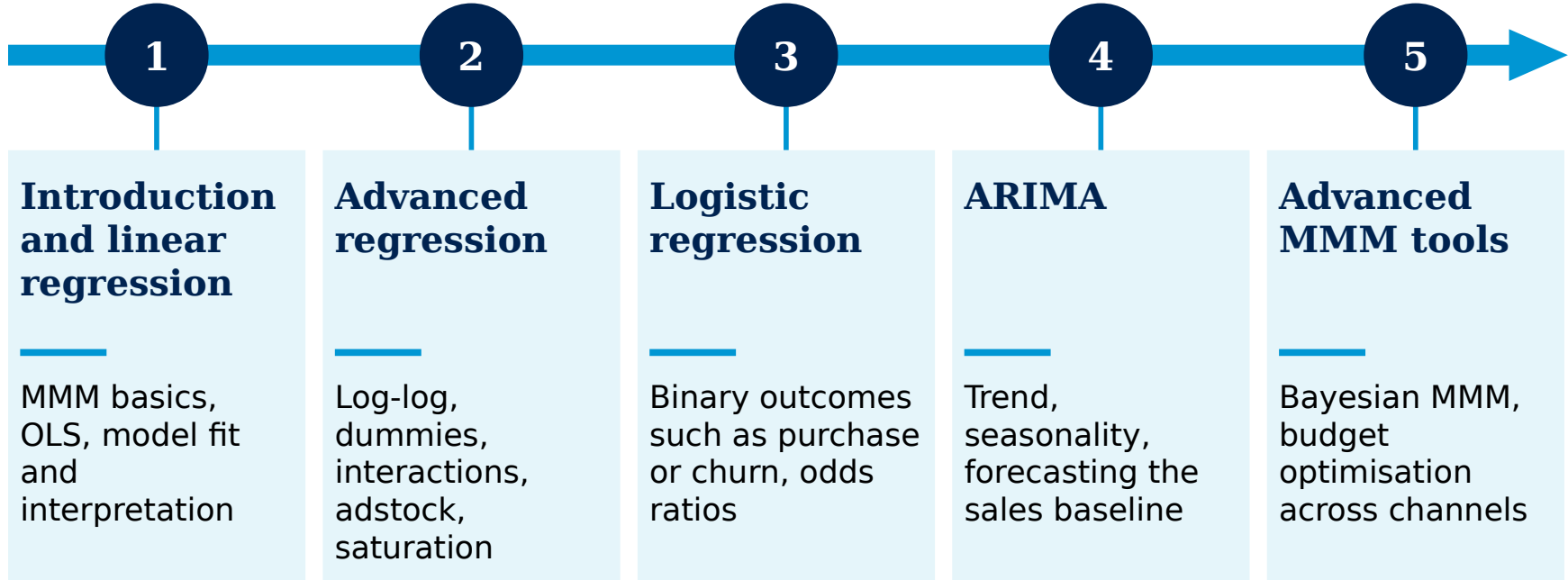


Tools: Python in Positron, GitHub for versioning and DataCamp courses; AI assistants do most of the typing, you check the results.



Case studies: Zalando and NEOH, plus a six-country teaching dataset with known answers for the labs.

Course roadmap: five sessions



Along the way: DataCamp certificates, coding exercises, guest speaker



At the end: written exam

Learning outcomes

On successful completion of this module, students will be able to:



Understand

- 1 **Explain the fundamentals of MMM**, its strengths and limits
- 2 **Describe real-world applications** of MMM in marketing



Apply

- 3 **Apply MMM in Python** to cross-sectional and longitudinal data
- 4 **Interpret results** and turn them into marketing decisions



Transfer

- 5 **Use AI tools** for analysis and reporting, and check the output
- 6 **Think critically** and work effectively in teams

Methods of teaching and learning

Every session

Concept lecture
60 min

Guided lab
90 min

Team exercise
60 min

Online quiz
not graded

Between sessions



- Read chapters and materials
- DataCamp courses and certificates
- Coding exercises
- Case study report in your group
- Q&A forum on Canvas



In the classroom



- Five in-person sessions
- Laptop with Python, Positron, GitHub
- AI-assisted coding, checked by you
- Guest speaker from industry



At the end: written exam, paper and pencil, multiple-choice questions.

Session	Topic	Methods
1	Introduction and linear regression	Course content and style; Python, Positron, GitHub and DataCamp; MMM fundamentals; OLS, model fit and interpretation
2	Advanced regression	Log-log elasticities, dummy variables, non-linear effects, moderation and mediation, adstock and saturation, diagnostics
3	Logistic regression	Binary outcomes such as purchase or churn, odds ratios, model fit and classification
4	ARIMA	Trend, seasonality and autocorrelation; ARIMA and ARIMAX forecasts as the sales baseline
5	Advanced MMM tools	Bayesian MMM with PyMC-Marketing and Google Meridian; budget optimisation across channels and countries

Assessment strategy

20 %	20 %	30 %	30 %
 20 % Online certificates DataCamp, 4 × 5 % Individual	 20 % Coding exercises 4 × 5 % Individual	 30 % Written exam paper and pencil, multiple choice Individual	 30 % Case study report written report, groups of max. 4 Group

Grading scale (WU)

1 Excellent 90 % and above	2 Good 80 to 89 %	3 Satisfactory 70 to 79 %	4 Sufficient 60 to 69 %	5 Fail below 60 %
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Attendance: at least 80 % of sessions. **AI tools:** expected; you own every number you report.

Software and set-up



Python

the language
for all analyses



Positron

the editor
where you
write, run and
see results



Quarto

turns code and
text into
reports and
slides



GitHub

versions and
shares your
work



AI assistant

GitHub Copilot,
free with the
student plan

Set-up in five steps

**Install
Python (uv)**

**Install
Positron**

**Install
Quarto**

**GitHub:
clone the
repo**

**Run the
check**



About 45 minutes. Follow the setup guide on Canvas before Session 1 and bring any installation problems to the first session.

4

DataCamp courses. Python courses assigned on Canvas, each ending with a certificate.

5 %

Per certificate. Upload each certificate on Canvas by its deadline.

20 %

In total. Together the four certificates count one fifth of the final grade.



Deadlines: the date for each certificate is on Canvas. The courses build the Python basics you need for the labs, so start early.

Coding exercises

4

exercises during the course

5 %

per exercise

20 %

of the final grade

How it works

**Guided lab
in class**

**Apply the code
to a similar
task**

**Submit on
Canvas**

**Feedback
and grade**

Rules



- Submit your Python or Quarto file; it must run from top to bottom
- AI assistants are allowed; add a short prompt log at the end
- Individual work: discuss ideas, but write and check your own code

Written exam

30 %

of the final grade

1

exam at the end of the course

MC

multiple-choice questions

Format



- Paper and pencil
- Multiple-choice questions
- Individual, closed book

Content



- All five sessions
- Concepts and interpretation
- Reading model output, not coding

Preparation



- Online quiz after every session (not graded)
- Session materials and book chapters
- Q&A forum on Canvas

Case study: written report

1

written report per
group

max. 4

students per group

30 %

of the final grade

2 weeks

after the last
session: deadline

What you submit



Quarto file

your analysis and text
in one reproducible
document



Rendered report

the output of the
Quarto file; HTML
recommended



Data file

the data you
analysed, so the
report can be rerun

The task: apply marketing mix modelling to a real marketing case, interpret the results and recommend a decision to management.

Support: online consulting hours (registration needed).



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